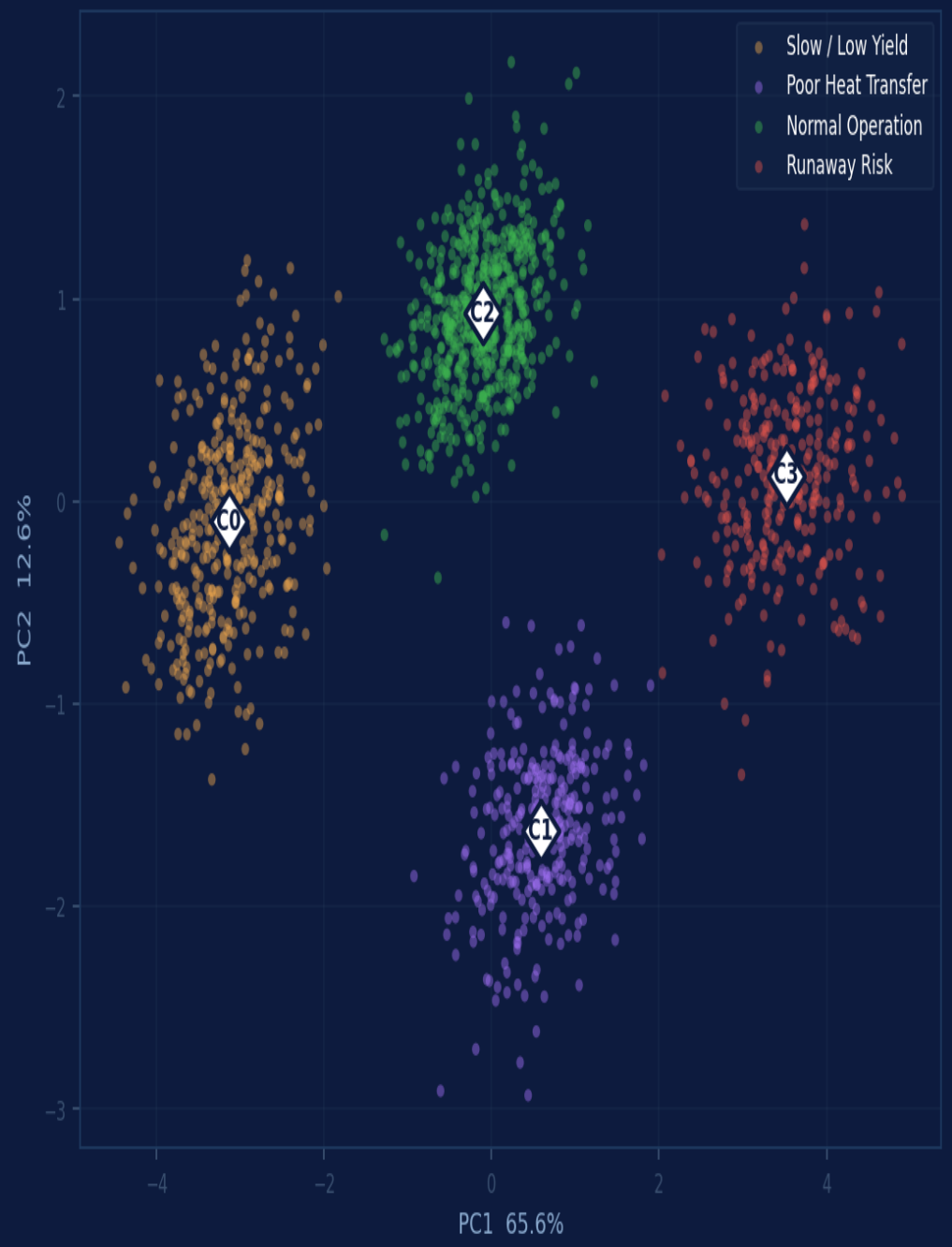
K-Means Clustering

Batch Reactor
Operating Mode Detection

"No alarm fired.
The data already knew."

K-Means · k=4 | StandardScaler | PCA · 2D | Hopkins · ARI



1,500
Batch Records

4
Operating Modes

1.000
ARI Score

0.803
Hopkins Statistic

The Real Problem

Threshold Blindness

Each sensor sits below its alarm cutoff. Temperature: 96°C (limit 105). Pressure: 3.8 bar (limit 5.0). Cooling: 52 L/min (above zero). No alarm fires — yet the batch is heading toward a runaway event.

The Combination Problem

A cooling flow of 52 L/min is fine at 80°C. It is catastrophically insufficient at 96°C. Process safety lives in the relationship between variables — exactly where single-threshold control is blind.

20% of Batches Are At Risk

Of 1,500 production runs, 300 batches (20%) fall into Cluster 3 — Aggressive / Runaway Risk. None of those batches triggered a hard alarm. All of them had a distinct multivariate signature.

K-Means sees all 8 process variables simultaneously — it lives in the same space where the risk forms.

Individual sensor readings can look safe while the process has already drifted into danger.

From Rules to Patterns

SENSOR-BY-SENSOR

IF temp > 105°C → ALARM

IF pressure > 5.0 bar → ALARM

IF cooling < 20 L/min → ALARM

→ Each threshold set independently

→ No variable interacts with another

→ Combination risk is invisible

→ Engineer writes rules from intuition

→ Rules must be revised after incidents

→

MULTIVARIATE SPACE

All 8 variables evaluated together

Euclidean distance in 8D feature space

K-Means finds natural groupings in data

→ 4 operating modes emerge from patterns

→ Combination risk becomes the signal

→ No expert labeling required

→ Centroids explain the mode in eng. units

→ Model improves as new batches arrive

The shift from threshold control to unsupervised pattern recognition.

The Data Behind the Problem

DCS

Distributed Control System

temp_max_c · pressure_max_bar

Thermocouple + pressure transmitter reading peak values per batch

VFD

Variable Frequency Drive

agitation_rpm

Agitator tachometer feedback — measures actual stirrer speed during reaction

MES

Manufacturing Execution System

conc_a_initial_pct · conc_b_initial_pct · reaction_time_min

Batch recipe dosing records and start/stop execution timestamps

PAT

Process Analytical Technology / LIMS

cooling_flow_l_min · final_conversion_pct

Magnetic flow meter on cooling jacket line · In-line NIR or GC end-of-batch yield

⚠ cluster_true is the ground-truth label — held out entirely during training. K-Means sees only the 8 process variables.

Why K-Means?

"The first question is not what threshold to set. It is: how many distinct modes does this reactor actually operate in?"

No Labels Required

The process historian contains no 'runaway' column. K-Means discovers structure from raw process data — no annotation, no historical incident reports needed.

Interpretable Centroids

Each cluster centroid in original engineering units (°C, bar, L/min) is a direct conversation with the process engineer. No black box — just a profile they already understand.

Deterministic Solution

With n_init=20, the algorithm runs 20 initializations and keeps the best. Stability std = 0.000 across 10 seeds: the same 4 modes emerge every time — critical for regulatory traceability.

Hopkins = 0.803 — Structure Is Real

Not all datasets cluster meaningfully. Hopkins > 0.75 confirms the batch data has genuine cluster tendency. We tested before committing — the geometry earned the algorithm.

Before you can write a good rule, you need to know what the operating modes actually look like.

Model Results

Operating Mode Profiles

Mode	Batches	Peak Temp	Cooling	Conversion
C0 · Slow / Low Yield	375 (25%)	68°C	101 L/min	85%
C1 · Poor Heat Transfer	300 (20%)	92°C	44 L/min	90%
C2 · Normal Operation	525 (35%)	80°C	90 L/min	95%
C3 · Runaway Risk	300 (20%)	100°C	60 L/min	98%

Validation Dashboard

Silhouette Score 0.440

Well-separated modes

Calinski-Harabasz 1435

Strong inter-cluster separation

Davies-Bouldin 0.917

< 1.0 — compact clusters

Hopkins Statistic 0.803

> 0.75 — genuine structure

ARI vs Ground Truth 1.000

Perfect recovery

K-Means fully recovered all 4 operating modes · ARI = 1.000 · Stability std = 0.000

From Insight to Action

C0

Slow Reaction / Low Yield

Trigger: Peak temp < 72°C · Cooling > 95 L/min · Conversion < 87%

→ ↑ Reactor temperature setpoint · Reduce cooling flow to release heat · Check reactant dosing sequence and agitator speed

C1

Poor Heat Transfer

Trigger: Temp 88–95°C · Cooling < 50 L/min · Pressure rising

→ Inspect cooling jacket for fouling · Verify cooling water supply pressure · Schedule heat exchanger maintenance — this cluster migrates to C3 without intervention

C2

Normal Operation

Trigger: Temp 76–84°C · Cooling 85–95 L/min · Conversion ≥ 93%

→ No corrective action. Use this batch as SPC baseline. Log centroid values as reference for the next production campaign.

C3

🚨 Runaway Risk — Immediate Intervention

Trigger: Temp > 96°C · Pressure > 4 bar · Cooling < 65 L/min · Fast batch (< 65 min)

→ **↑ Cooling flow to maximum · Halt or reduce reactant dosing · Verify emergency vent and interlock status · Notify process engineer and safety officer immediately**

The centroid is the diagnosis. The playbook is the prescription.

Batch Classifier — What-If Scenarios

A — Baseline	
Temp (max)	80.0°C
Pressure	2.5 bar
Agitation	300 rpm
Conc A	35%
Conc B	35%
Cooling	90 L/min
Rxn Time	80 min
Conversion	95%

B — Runaway Alert	
Temp (max)	102°C
Pressure	4.8 bar
Agitation	390 rpm
Conc A	42%
Conc B	42%
Cooling	55 L/min
Rxn Time	58 min
Conversion	98.5%

C — Cooling Failure	
Temp (max)	93°C
Pressure	3.6 bar
Agitation	282 rpm
Conc A	34%
Conc B	34%
Cooling	**40 L/min**
Rxn Time	87 min
Conversion	90%

✅ **Normal Operation**

Anomaly Index: 4%

🚨 **Runaway Risk**

Anomaly Index: 97%

⚠ **Poor Heat Transfer**

Anomaly Index: 71%

Scenario C: reducing cooling from 90 → 40 L/min alone moves the batch from C2 (Normal) to C1 (Poor Heat Transfer) — without triggering any hard alarm.

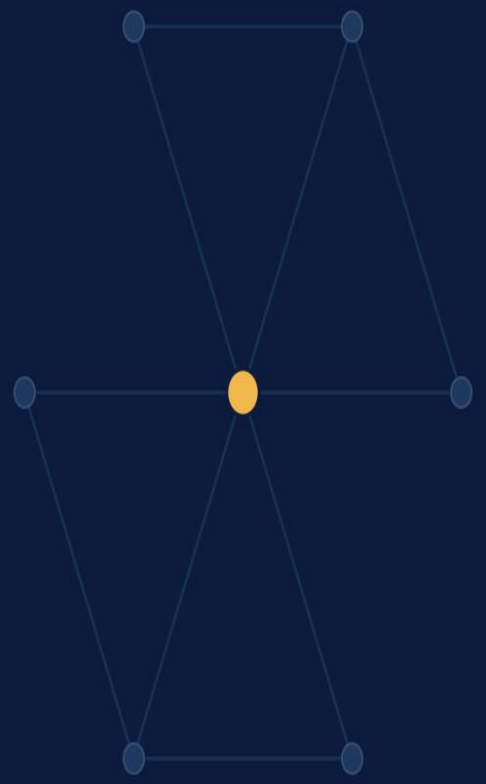
Enter any 8 process readings → the model returns the operating mode and recommended actions.

The reactor doesn't announce what it's about to do.

Neither does the data — until you listen to all of it at once.

Eight variables. Four modes. Zero labels. One pattern that threshold alarms could never see.

Download the notebook. Run it on your own process historian data.

Change k. Change the scaler. Challenge every centroid.

Because unsupervised learning has no finish line.